MindTrack/Brain-Tasks-App

Docker:

Creating Docker File:

```
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ vi Dockerfile

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ cat Dockerfile
FROM nginx:alpine

# Remove default nginx static files
RUN rm -rf /usr/share/nginx/html/*

# Copy app files
COPY . /usr/share/nginx/html

# Expose port
EXPOSE 3000

# Update nginx config to use port 3000
RUN sed -i 's/80/3000/g' /etc/nginx/conf.d/default.conf

CMD ["nginx", "-g", "daemon off;"]

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$
```

Building the Application using Dockerfile:

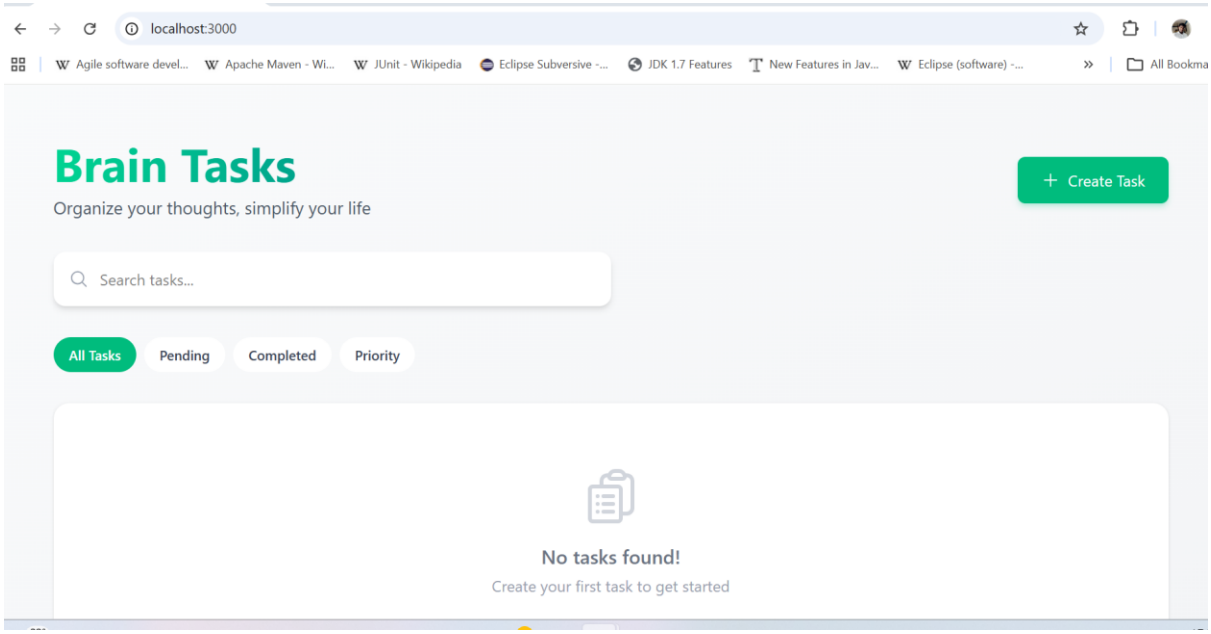
```
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ ls
Dockerfile  assets/  index.html  vite.svg

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ docker build -t brain-tasks-app .
[+] Building 110.4s (9/9) FINISHED                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                0.1s
 => => transferring dockerfile: 335B                                                                                                0.0s
 => [internal] load metadata for docker.io/library/nginx:alpine                                                                     8.3s
 => [internal] load .dockerignore                                                                                                   0.1s
 => => transferring context: 2B                                                                                                     0.0s
 => [1/4] FROM docker.io/library/nginx:alpine@sha256:e7257f1ef28ba17cf7c248cb8ccf6f0c6e0228ab9c315c152f9c203cd34cf6d1              98.5s
 => => resolve docker.io/library/nginx:alpine@sha256:e7257f1ef28ba17cf7c248cb8ccf6f0c6e0228ab9c315c152f9c203cd34cf6d1               0.1s
 => => sha256:3370263bc02adcf5c4f51831d2bf1d54dbf9a6a80b0bf32c5c9b9400630eaa08 20.25MB / 20.25MB                                  97.2s
 => => sha256:4d9d41f3822d171ccc5f2cdfd75ad846ac4c7ed1cd36fb998fe2c0ce4501647b 1.40kB / 1.40kB                                     1.3s
 => => sha256:c2bd5ab17727idd59f19a46c214b1327f5c428cd075437ec0155ae71d0cdadc1 1.21kB / 1.21kB                                     2.3s
 => => sha256:3f4ad4352d4f91018e2b4910b9db24c08e70192c3b75d0d6fff0120c838aa0bb 402B / 402B                                         1.4s
 => => sha256:cf1159c696ee2a72b85634360dbada071db61bceaad253db7fda65c45a58414c 953B / 953B                                         0.7s
 => => sha256:91d1c9c22f2c631288354fadb2decc448ce151d7a197c167b206588e09dcd50a 626B / 626B                                         0.7s
 => => sha256:8892f80f46a05d59a4cde3bcbb1dd26ed2441d4214870a4a7b318eaa476a0a54 1.87MB / 1.87MB                                     6.9s
 => => sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153 3.86MB / 3.86MB                                    12.1s
 => => extracting sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153                                           0.2s
 => => extracting sha256:8892f80f46a05d59a4cde3bcbb1dd26ed2441d4214870a4a7b318eaa476a0a54                                           0.2s
 => => extracting sha256:91d1c9c22f2c631288354fadb2decc448ce151d7a197c167b206588e09dcd50a                                           0.0s
 => => extracting sha256:cf1159c696ee2a72b85634360dbada071db61bceaad253db7fda65c45a58414c                                           0.0s
 => => extracting sha256:3f4ad4352d4f91018e2b4910b9db24c08e70192c3b75d0d6fff0120c838aa0bb                                           0.0s
 => => extracting sha256:c2bd5ab17727idd59f19a46c214b1327f5c428cd075437ec0155ae71d0cdadc1                                           0.0s
 => => extracting sha256:4d9d41f3822d171ccc5f2cdfd75ad846ac4c7ed1cd36fb998fe2c0ce4501647b                                           0.1s
 => => extracting sha256:3370263bc02adcf5c4f51831d2bf1d54dbf9a6a80b0bf32c5c9b9400630eaa08                                           0.5s
 => [internal] load build context                                                                                                   0.9s
 => => transferring context: 318.26kB                                                                                               0.7s
 => [2/4] RUN rm -rf /usr/share/nginx/html/*                                                                                        1.2s
 => [3/4] COPY . /usr/share/nginx/html                                                                                              0.2s
 => [4/4] RUN sed -i 's/80/3000/g' /etc/nginx/conf.d/default.conf                                                                   0.4s
 => exporting to image                                                                                                              1.1s
 => => exporting layers                                                                                                             0.6s
 => => exporting manifest sha256:0204efb37b35691473f35d6c953e9d65b80dd8795d8e7dbc71214bf8f61d5450                                   0.0s
 => => exporting config sha256:886f7f1eefbf1fea79c9f8c2b8ba845130a942557e80d9acdc1c569bd0d06a80                                    0.0s
 => => exporting attestation manifest sha256:da317c53c3c24d4350f4d2ec3a4f68635a0485e4cf73393efa84201dfa790dc4                       0.1s
 => => exporting manifest list sha256:af50fcd54a4ffa974a642313acc7068536f78314383d5fcfc96f700ca587dac9                              0.0s
 => => naming to docker.io/library/brain-tasks-app:latest                                                                           0.0s
 => => unpacking to docker.io/library/brain-tasks-app:latest                                                                        0.1s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/6nzr5jrijhb5u95icor3a0wtb
```

```
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ docker images
                                                                                                                                                                    i Info →
IMAGE                                                                                                                                                ID             DISK USAGE   CONTENT SIZE   EXTRA
brain-tasks-app:latest                                                                                                                               af50fcd54a4f       93.1MB         26.1MB
gcr.io/k8s-minikube/kicbase:v0.0.49                                                                                                                  c7ddf4818024       1.91GB          539MB
gcr.io/k8s-minikube/kicbase@sha256:e6daddbb1dc09ccd195c5605f65e2d38406c36ef36c5a492ffe805d9d36f4945                                                  e6daddbb1dc0       1.91GB          539MB    U

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$
```

```
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ docker run -d -p 3000:3000 brain-tasks-app:latest
cad923cf21415e3b80162c583d146f97b9609deb9b50a5ce4575b525194a2864

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ docker ps
CONTAINER ID   IMAGE                   COMMAND                  CREATED          STATUS          PORTS                                       NAMES
cad923cf2141   brain-tasks-app:latest  "/docker-entrypoint.…"   11 seconds ago   Up 9 seconds    0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp   clever_beaver

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$
```

```
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ curl http://localhost:3000
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="https://is1-ssl.mzstatic.com/image/thumb/Purple221/v4/1b/a6/70/1ba67006-9749-9567-151b-731d5cba4f93/AppIcon-0-0-1x_U007epad-
0-1-0-85-220.jpeg/512x512bb.jpg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Brain Task</title>
    <script type="module" crossorigin src="/assets/index-BHGiHu50.js"></script>
    <link rel="stylesheet" crossorigin href="/assets/index-DPTLVrPB.css">
  </head>
  <body>
    <div id="root">
  </body>
</html>
```

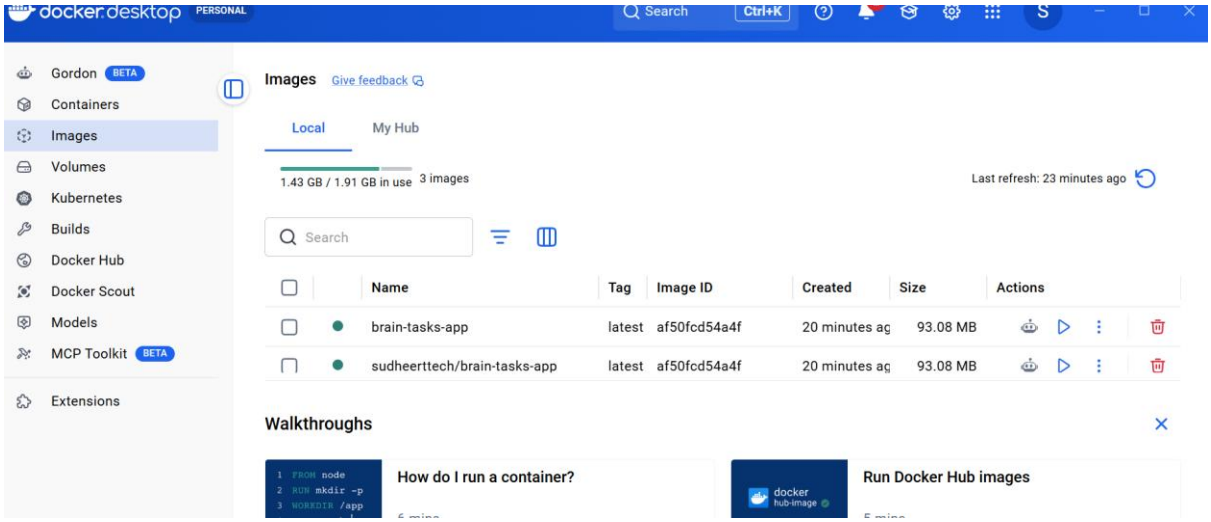
Output:



Docker registry:

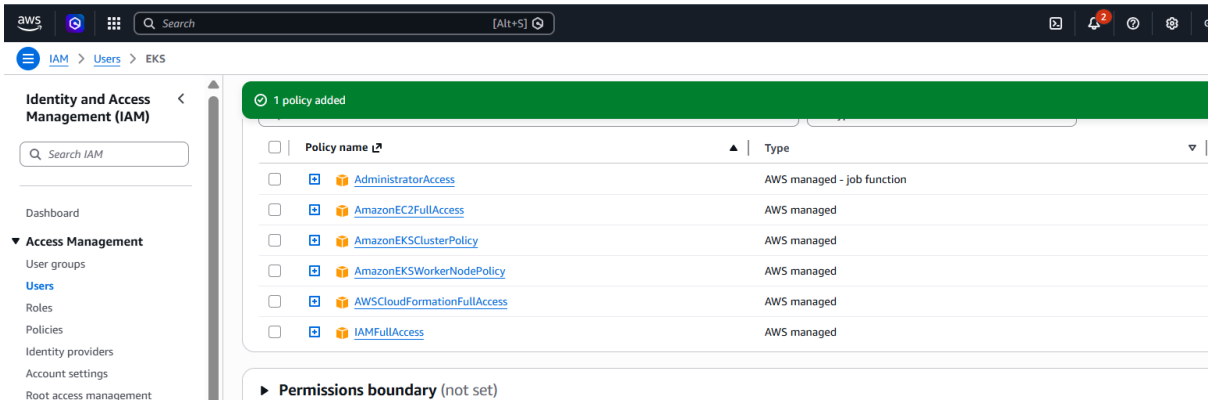
Pushed docker Image to Dockerhub

```
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ docker login
Authenticating with existing credentials... [Username: sudheerttech]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ docker tag brain-tasks-app sudheerttech/brain-tasks-app:latest

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ docker push sudheerttech/brain-tasks-app:latest
The push refers to repository [docker.io/sudheerttech/brain-tasks-app]
91d1c9c22f2c: Pushed
589002ba0eae: Pushed
3f4ad4352d4f: Pushed
a9e20c319e81: Pushed
8892f80f46a0: Pushed
98a1018da615: Pushed
cf1159c696ee: Pushed
049e410d88e1: Pushed
4d9d41f3822d: Pushed
018b60290207: Pushed
c2bd5ab17727: Pushed
3370263bc02a: Pushed
latest: digest: sha256:af50fcd54a4ffa974a642313acc7068536f78314383d5fcfc96f700ca587dac9 size: 856

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
```



Kubernetes:

Added required policies to IAM role:



Created AWS EKS cluster:

eksctl create cluster --name brain-tasks-cluster --region us-east-1 --version 1.32 --nodegroup-name standard-workers --node-type t3.small --nodes 1 --nodes-min 1 --nodes-max 1 --managed

```
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ eksctl create cluster \
 --name brain-tasks-cluster \
 --region us-east-1 \
 --version 1.32 \
 --nodegroup-name standard-workers \
 --node-type t3.medium \
 --nodes 1 \
 --nodes-min 1 \
 --nodes-max 1 \
 --managed
2026-03-30 18:53:47 [ℹ]  eksctl version 0.221.0
2026-03-30 18:53:47 [ℹ]  using region us-east-1
2026-03-30 18:53:55 [ℹ]  setting availability zones to [us-east-1b us-east-1f]
2026-03-30 18:53:55 [ℹ]  subnets for us-east-1b - public:192.168.0.0/19 private:192.168.64.0/19
2026-03-30 18:53:55 [ℹ]  subnets for us-east-1f - public:192.168.32.0/19 private:192.168.96.0/19
2026-03-30 18:53:55 [ℹ]  nodegroup "standard-workers" will use "" [AmazonLinux2023/1.32]
2026-03-30 18:53:55 [!]  Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons will no
 be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: https://eksctl.io/usa
-mode/
2026-03-30 18:53:55 [ℹ]  using Kubernetes version 1.32
2026-03-30 18:53:55 [ℹ]  creating EKS cluster "brain-tasks-cluster" in "us-east-1" region with managed nodes
2026-03-30 18:53:55 [ℹ]  will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-03-30 18:53:55 [ℹ]  if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-east-1 --cluster=brain-tasks-clus
2026-03-30 18:53:55 [ℹ]  Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "brain-tasks-cluster" in "us-east-1"
2026-03-30 18:53:55 [ℹ]  CloudWatch logging will not be enabled for cluster "brain-tasks-cluster" in "us-east-1"
2026-03-30 18:53:55 [ℹ]  you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=us-east-1 --clu
ain-tasks-cluster'
2026-03-30 18:53:55 [ℹ]  default addons vpc-cni, kube-proxy, coredns, metrics-server were not specified, will install them as EKS addons
2026-03-30 18:53:55 [ℹ]
2 sequential tasks: { create cluster control plane "brain-tasks-cluster",
    2 sequential sub-tasks: {
```

```
2026-03-30 20:06:47 [ℹ]  nodegroup "standard-workers" has 1 node(s)
2026-03-30 20:06:47 [ℹ]  node "ip-192-168-53-111.ec2.internal" is ready
2026-03-30 20:06:47 [ℹ]  waiting for at least 1 node(s) to become ready in "standard-workers"
2026-03-30 20:06:47 [ℹ]  nodegroup "standard-workers" has 1 node(s)
2026-03-30 20:06:47 [ℹ]  node "ip-192-168-53-111.ec2.internal" is ready
2026-03-30 20:06:47 [✓]  created 1 managed nodegroup(s) in cluster "brain-tasks-cluster"
2026-03-30 20:06:50 [ℹ]  creating addon: metrics-server
2026-03-30 20:06:50 [ℹ]  successfully created addon: metrics-server
2026-03-30 20:06:54 [ℹ]  kubectl command should work with "C:\\Users\\hp\\.kube\\config", try 'kubectl get node
2026-03-30 20:06:54 [✓]  EKS cluster "brain-tasks-cluster" in "us-east-1" region is ready

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ eksctl get cluster
NAME                    REGION          EKSCTL CREATED
brain-tasks-cluster     us-east-1       True

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ kubectl get nodes
NAME                                   STATUS   ROLES    AGE     VERSION
ip-192-168-53-111.ec2.internal         Ready    <none>   4m25s   v1.32.12-eks-f69f56f

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$
```

Creating **deployment.yaml** and **service.yaml**:

deployment.yaml

---

apiVersion: apps/v1

kind: Deployment

metadata:

  name: brain-tasks-app

spec:

  replicas: 2

  selector:

    matchLabels:

```
    app: brain-tasks
  template:
    metadata:
      labels:
        app: brain-tasks
    spec:
      containers:
      - name: brain-tasks-container
        image: sudheerttech/brain-tasks-app
        ports:
        - containerPort: 3000
```

service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: brain-tasks-service
spec:
  type: LoadBalancer
  selector:
    app: brain-tasks
  ports:
    - protocol: TCP
      port: 80
      targetPort: 3000
```

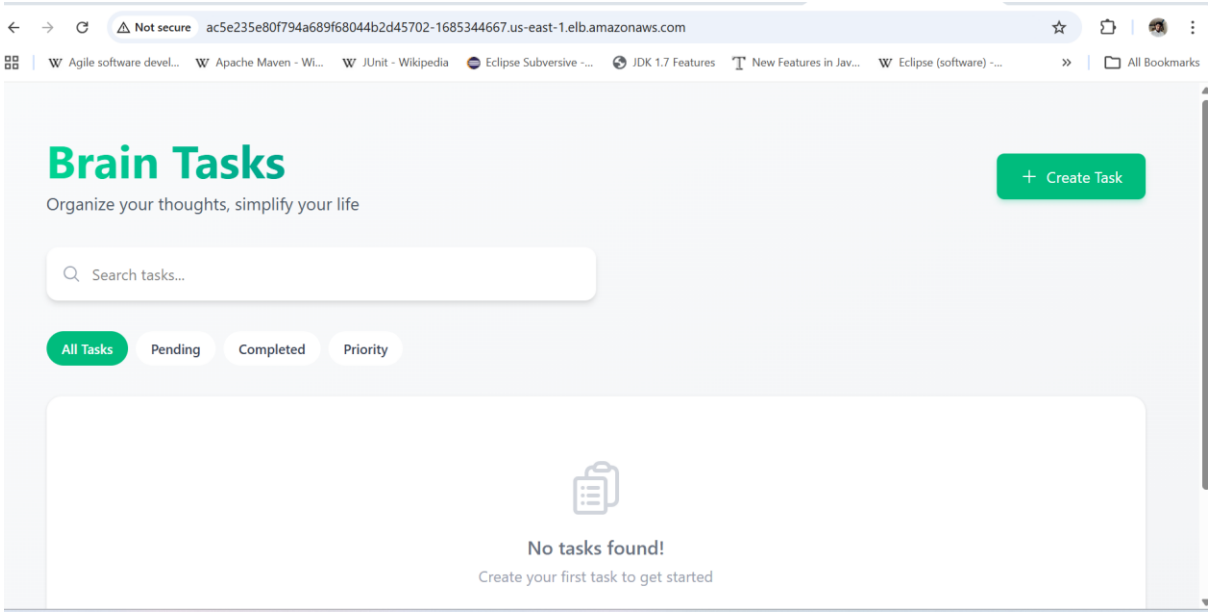
```
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ eksctl get cluster
NAME                    REGION          EKSCTL CREATED
brain-tasks-cluster     us-east-1       True

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ kubectl get nodes
NAME                              STATUS   ROLES    AGE     VERSION
ip-192-168-53-111.ec2.internal    Ready    <none>   4m25s   v1.32.12-eks-f69f56f

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ vi deployment.yaml

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ vi service.yaml

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: brain-tasks-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: brain-tasks
  template:
    metadata:
      labels:
        app: brain-tasks
    spec:
      containers:
      - name: brain-tasks-container
        image: sudheerttech/brain-tasks-app
        ports:
        - containerPort: 3000

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: brain-tasks-service
spec:
  type: LoadBalancer
  selector:
    app: brain-tasks
  ports:
    - protocol: TCP
      port: 80
      targetPort: 3000
```

Deploying the .yaml files :

```
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ kubectl apply -f deployment.yaml
E0330 20:21:20.824025   29820 request.go:1196] "Unexpected error when reading response body" err="context deadline exceeded (Client.Timeout or context cancellation while re
ading body)"
deployment.apps/brain-tasks-app created

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ AC

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ kubectl get deployments
NAME              READY   UP-TO-DATE   AVAILABLE   AGE
brain-tasks-app   2/2     2            2           69s

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ kubectl describe deployment brain-tasks-app
Name:                   brain-tasks-app
Namespace:              default
CreationTimestamp:      Mon, 30 Mar 2026 20:21:42 +0530
Labels:                 <none>
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=brain-tasks
Replicas:               2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=brain-tasks
  Containers:
   brain-tasks-container:
    Image:        sudheerttech/brain-tasks-app
    Port:         3000/TCP
    Host Port:    0/TCP
    Environment:  <none>
    Mounts:       <none>
  Volumes:        <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   brain-tasks-app-6dc9bc6895 (2/2 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  89s   deployment-controller  Scaled up replica set brain-tasks-app-6dc9bc6895 from 0 to 2
```

```
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE     IP               NODE                                       NOMINATED NODE   READINESS GATES
brain-tasks-app-6dc9bc6895-m5vsb    1/1     Running   0          2m45s   192.168.59.65    ip-192-168-53-111.ec2.internal   <none>           <none>
brain-tasks-app-6dc9bc6895-vlng4    1/1     Running   0          2m45s   192.168.36.52    ip-192-168-53-111.ec2.internal   <none>           <none>

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ kubectl apply -f service.yaml
E0330 20:26:03.936808   10860 request.go:1196] "Unexpected error when reading response body" err="net/http: request canceled (Client.Timeout or context cancellation while r
eading body)"
service/brain-tasks-service created

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ kubectl get svc
NAME                 TYPE           CLUSTER-IP      EXTERNAL-IP                                                              PORT(S)        AGE
brain-tasks-service  LoadBalancer   10.100.77.222   ac5e235e80f794a689f68044b2d45702-1685344667.us-east-1.elb.amazonaws.com  80:30622/TCP   3m58s
kubernetes           ClusterIP      10.100.0.1      <none>                                                                   443/TCP        31m

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ kubectl describe svc brain-tasks-service
Name:                     brain-tasks-service
Namespace:                default
Labels:                   <none>
Annotations:              <none>
Selector:                 app=brain-tasks
Type:                     LoadBalancer
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.100.77.222
IPs:                      10.100.77.222
LoadBalancer Ingress:     ac5e235e80f794a689f68044b2d45702-1685344667.us-east-1.elb.amazonaws.com
Port:                     <unset>  80/TCP
TargetPort:               3000/TCP
NodePort:                 <unset>  30622/TCP
Endpoints:                192.168.36.52:3000,192.168.59.65:3000
Session Affinity:         None
External Traffic Policy:  Cluster
Internal Traffic Policy:  Cluster
Events:
  Type    Reason                Age    From                Message
  ----    ------                ----   ----                -------
  Normal  EnsuringLoadBalancer  4m22s  service-controller  Ensuring load balancer
  Normal  EnsuredLoadBalancer   4m18s  service-controller  Ensured load balancer

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
```

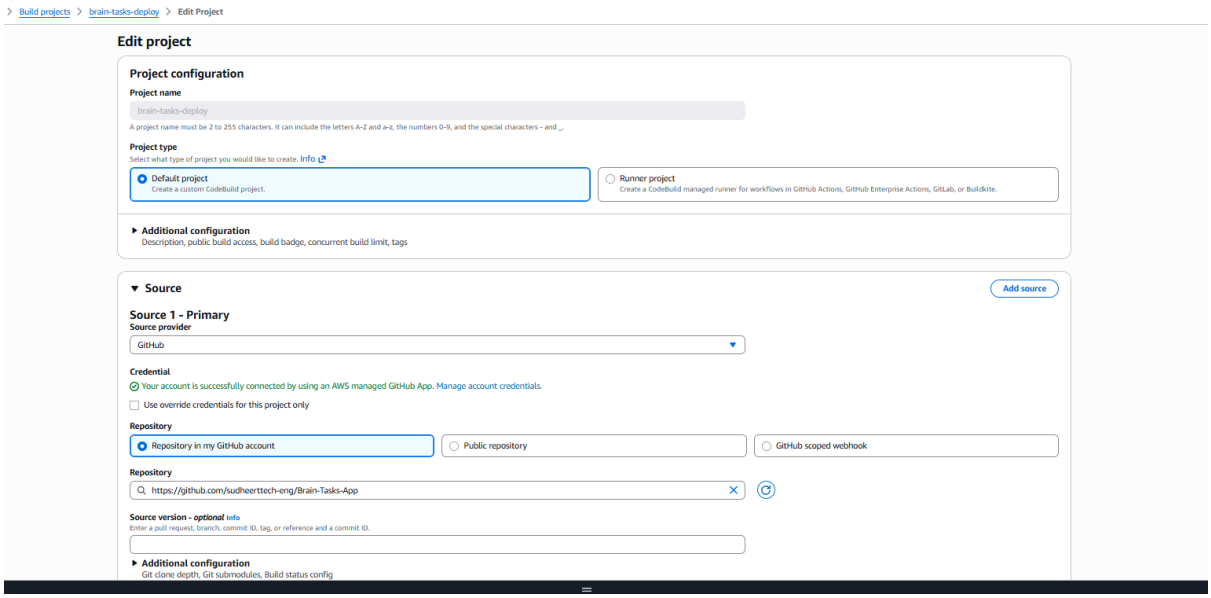
Accessing the app by using external-IP of the service:

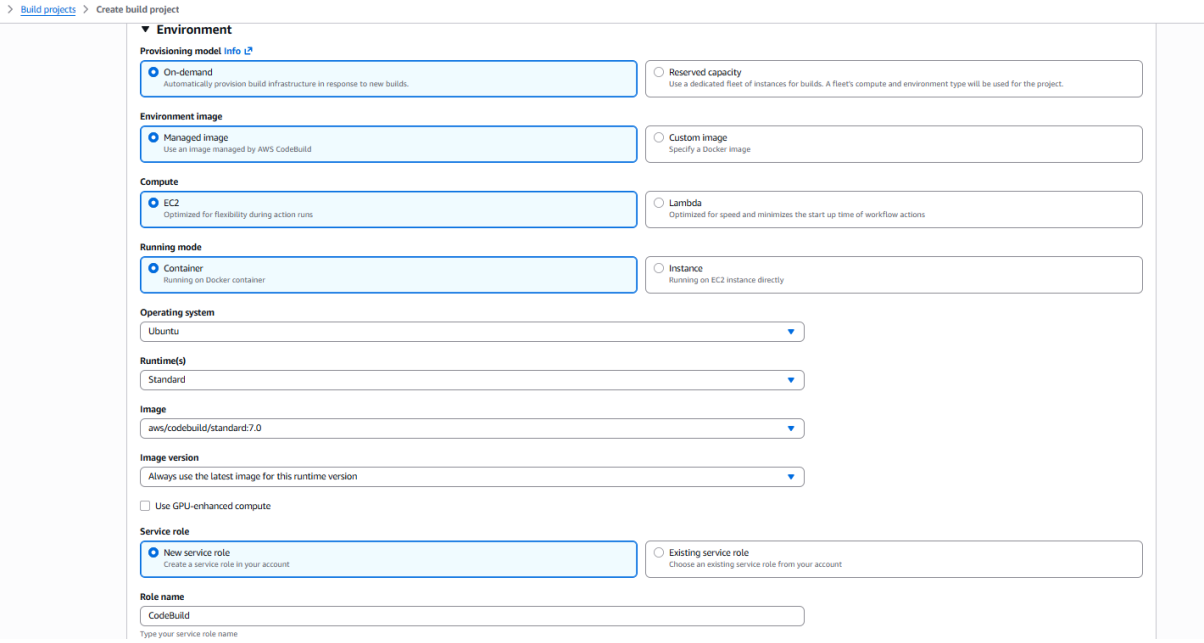


Deploy using kubectl via CodeBuild or Codepipeline.

CodeBuild Setup:

Creating code build project and Connected to Github repository:





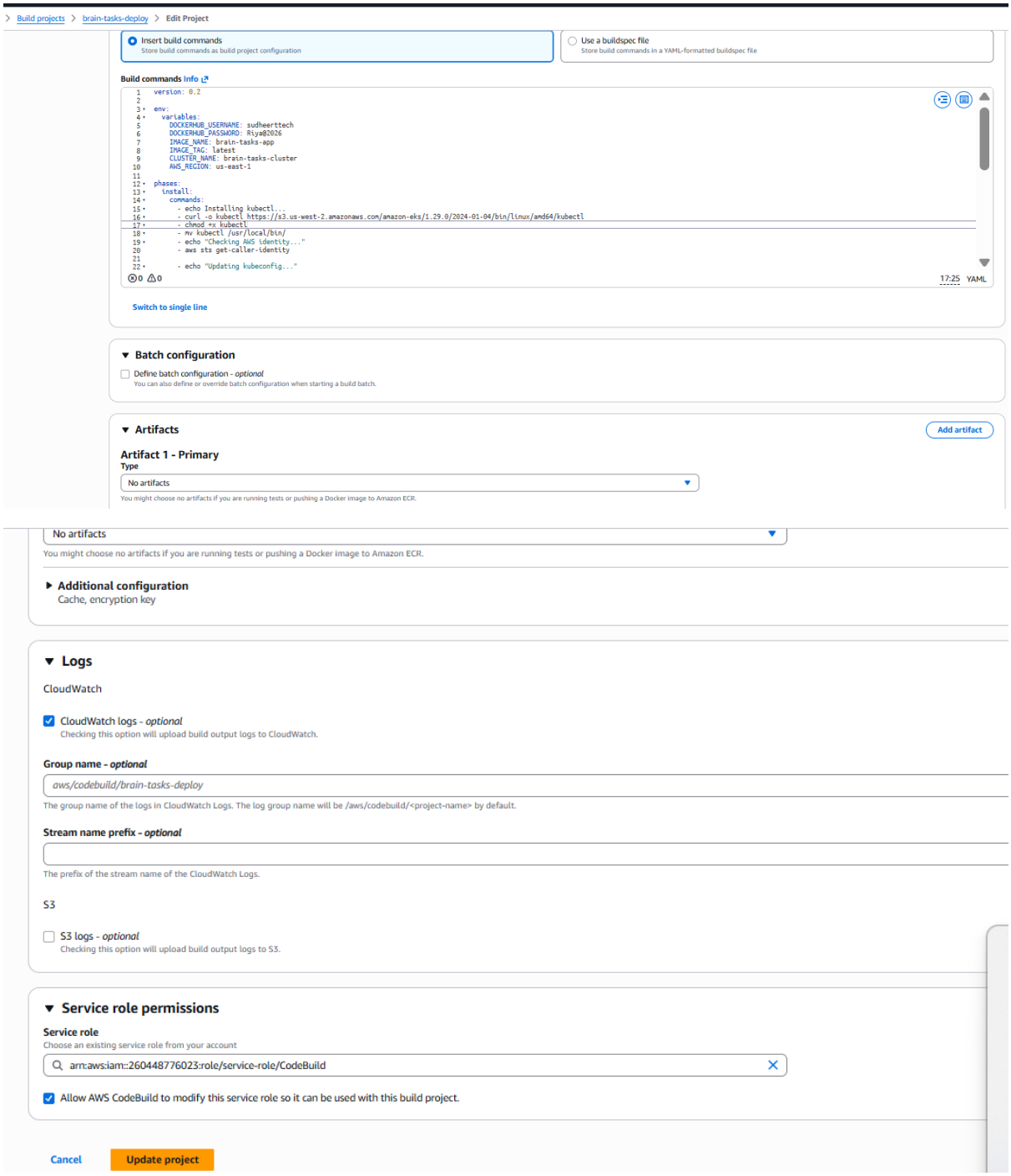
Created buildspec.yml:

```
$ cat buildspec.yml
version: 0.2

env:
  variables:
    DOCKERHUB_USERNAME: sudheerttech
    DOCKERHUB_PASSWORD: Riya@2026
    IMAGE_NAME: brain-tasks-app
    IMAGE_TAG: latest
    CLUSTER_NAME: brain-tasks-cluster
    AWS_REGION: us-east-1

phases:
  install:
    commands:
      - echo Installing kubectl...
      - curl -o kubectl https://s3.us-west-2.amazonaws.com/amazon-eks/1.29.0/2024-01-04/bin/linux/amd64/kubectl
      - chmod +x kubectl
      - mv kubectl /usr/local/bin/
      - echo "Checking AWS identity..."
      - aws sts get-caller-identity

      - echo "Updating kubeconfig..."
      - aws eks --region $AWS_REGION update-kubeconfig --name $CLUSTER_NAME

  pre_build:
    commands:
      - echo Logging into Docker Hub...
      - echo $DOCKERHUB_PASSWORD | docker login -u $DOCKERHUB_USERNAME --password-stdin
      - echo Build started on `date`
      - docker build -t $IMAGE_NAME:$IMAGE_TAG .
      - docker tag $IMAGE_NAME:$IMAGE_TAG $DOCKERHUB_USERNAME/$IMAGE_NAME:$IMAGE_TAG
      - aws eks --region $AWS_REGION update-kubeconfig --name $CLUSTER_NAME

  build:
    commands:
      - echo Pushing Docker image to Docker Hub...
      - docker push $DOCKERHUB_USERNAME/$IMAGE_NAME:$IMAGE_TAG
      - echo Deploying to EKS cluster...
      - kubectl apply -f deployment.yaml
      - kubectl apply -f service.yaml
      - kubectl set image deployment/brain-tasks-app brain-tasks-container=$DOCKERHUB_USERNAME/$IMAGE_NAME:$IMAGE_TAG
      - kubectl rollout status deployment/brain-tasks-app

  post_build:
    commands:
      - echo Build and deployment complete on `date`

artifacts:
  files: []


hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$
```

buildspec.yml

```
version: 0.2


env:
  variables:
    DOCKERHUB_USERNAME: sudheerttech
    DOCKERHUB_PASSWORD: Riya@2026
    IMAGE_NAME: brain-tasks-app
    IMAGE_TAG: latest
    CLUSTER_NAME: brain-tasks-cluster
    AWS_REGION: us-east-1


phases:
  install:
    commands:
      - echo Installing kubectl...
      - curl -o kubectl https://s3.us-west-2.amazonaws.com/amazon-eks/1.29.0/2024-01-04/bin/linux/amd64/kubectl
      - chmod +x kubectl
      - mv kubectl /usr/local/bin/
      - echo "Checking AWS identity..."
      - aws sts get-caller-identity


      - echo "Updating kubeconfig..."
      - aws eks --region $AWS_REGION update-kubeconfig --name $CLUSTER_NAME


  pre_build:
    commands:
      - echo Logging into Docker Hub...
      - echo $DOCKERHUB_PASSWORD | docker login -u $DOCKERHUB_USERNAME --password-stdin
      - echo Build started on `date`
```

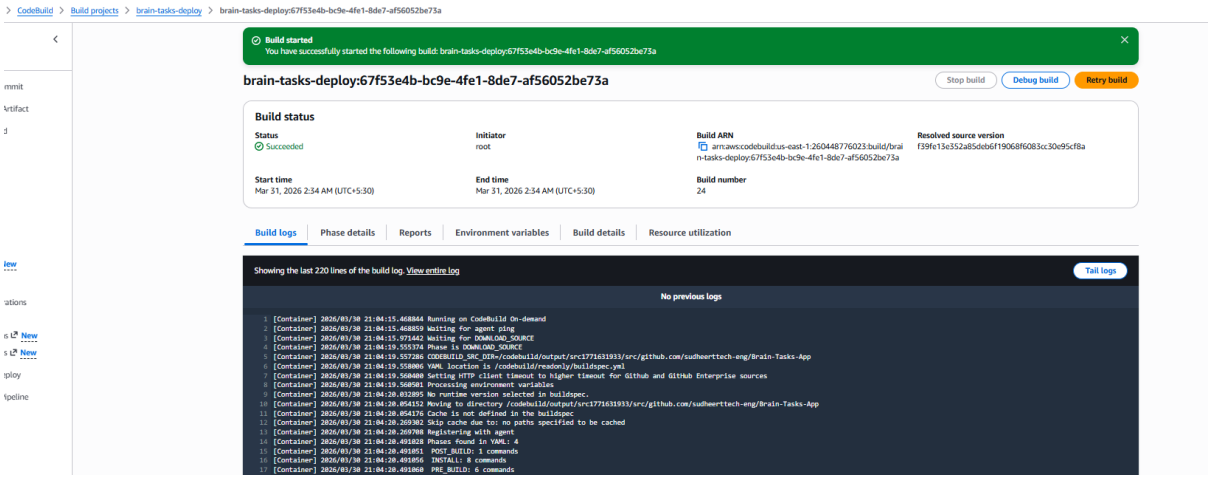
```
    - docker build -t $IMAGE_NAME:$IMAGE_TAG .
    - docker tag $IMAGE_NAME:$IMAGE_TAG
$DOCKERHUB_USERNAME/$IMAGE_NAME:$IMAGE_TAG
    - aws eks --region $AWS_REGION update-kubeconfig --name $CLUSTER_NAME


  build:
   commands:
    - echo Pushing Docker image to Docker Hub...
    - docker push $DOCKERHUB_USERNAME/$IMAGE_NAME:$IMAGE_TAG
    - echo Deploying to EKS cluster...
    - kubectl apply -f deployment.yaml
    - kubectl apply -f service.yaml
    - kubectl set image deployment/brain-tasks-app brain-tasks-
container=$DOCKERHUB_USERNAME/$IMAGE_NAME:$IMAGE_TAG
    - kubectl rollout status deployment/brain-tasks-app


  post_build:
   commands:
    - echo Build and deployment complete on `date`
artifacts:
  files: []
```



Create policy for the role of CodeBuild:

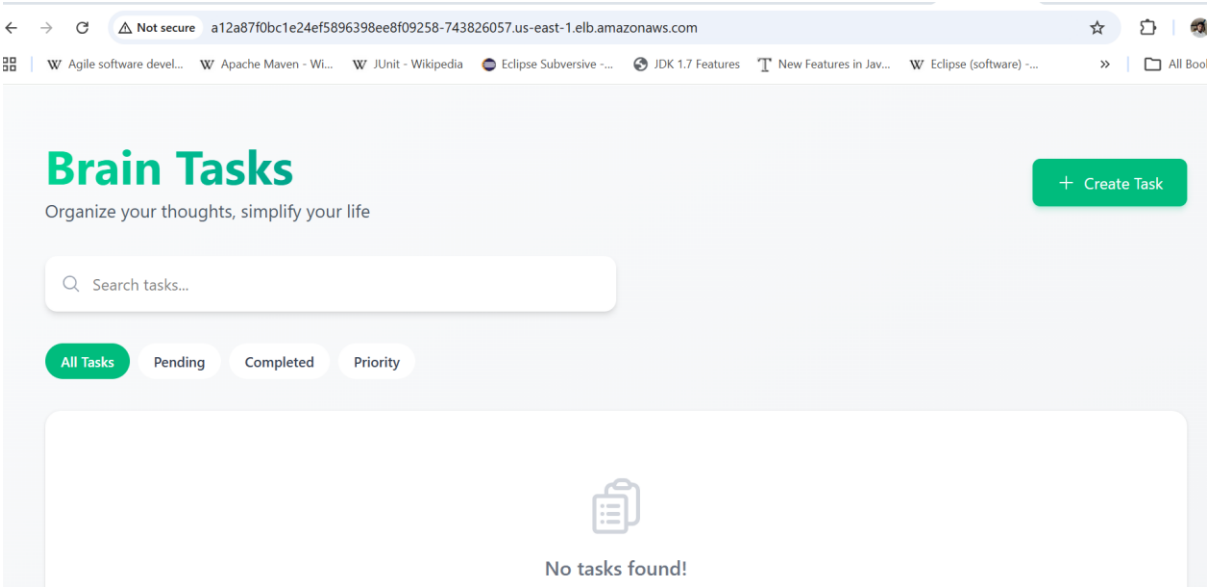
Add CodeBuild Role to EKS aws-auth ConfigMap: This gives eks:DescribeCluster permission, which is required for aws eks update-kubeconfig

```
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App (main)
$ kubectl edit configmap aws-auth -n kube-system
E0331 02:10:40.952171   16892 request.go:1196] "Unexpected error when reading response body" err="context deadline exceeded (Client.Timeout or context cancellation while reading body)"
configmap/aws-auth edited

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App (main)
$ kubectl get configmap aws-auth -n kube-system -o yaml
apiVersion: v1
data:
  mapRoles: |
    - rolearn: arn:aws:iam::260448776023:role/CodeBuild
      username: codebuild
      groups:
      - system:masters
    - rolearn: arn:aws:iam::260448776023:role/eksctl-brain-tasks-cluster-nodegro-NodeInstanceRole-hbJ4hj3iO09u
      groups:
      - system:bootstrappers
      - system:nodes
      username: system:node:{{EC2PrivateDNSName}}
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-30T14:34:27Z"
  name: aws-auth
  namespace: kube-system
  resourceVersion: "59530"
  uid: 6dc661ce-d21f-4c2d-aab8-13df634f59fa
```




```
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App (main)
$ kubectl get nodes
NAME                                       STATUS   ROLES    AGE     VERSION
ip-192-168-53-111.ec2.internal             Ready    <none>   6h58m   v1.32.12-eks-f69f56f

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App (main)
$ kubectl get deployments
NAME              READY   UP-TO-DATE   AVAILABLE   AGE
brain-tasks-app   2/2     2            2           28m

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App (main)
$ kubectl get svc
NAME                 TYPE           CLUSTER-IP     EXTERNAL-IP                                                              PORT(S)        AGE
brain-tasks-service  LoadBalancer   10.100.28.86   a12a87f0bc1e24ef5896398ee8f09258-743826057.us-east-1.elb.amazonaws.com   80:31598/TCP   28m
kubernetes           ClusterIP      10.100.0.1     <none>                                                                   443/TCP        5h59m

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App (main)
```

Load Balancer ARN: http://a12a87f0bc1e24ef5896398ee8f09258-743826057.us-east-1.elb.amazonaws.com/



Git repository:

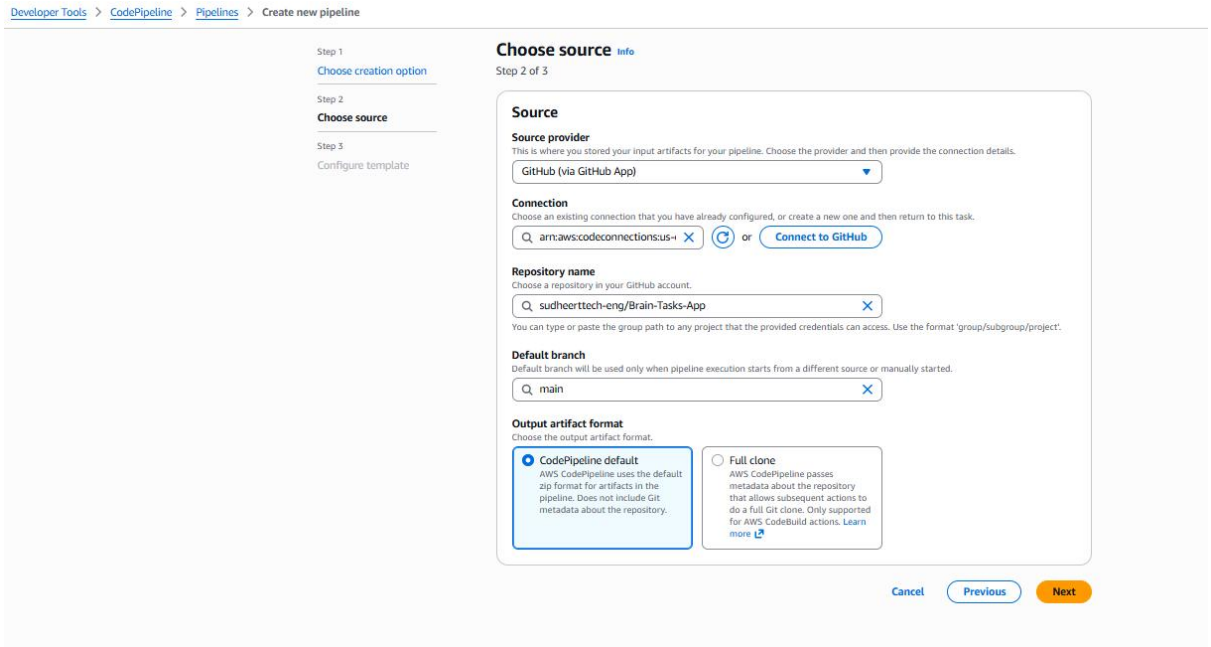
Pushed all files to the repository.

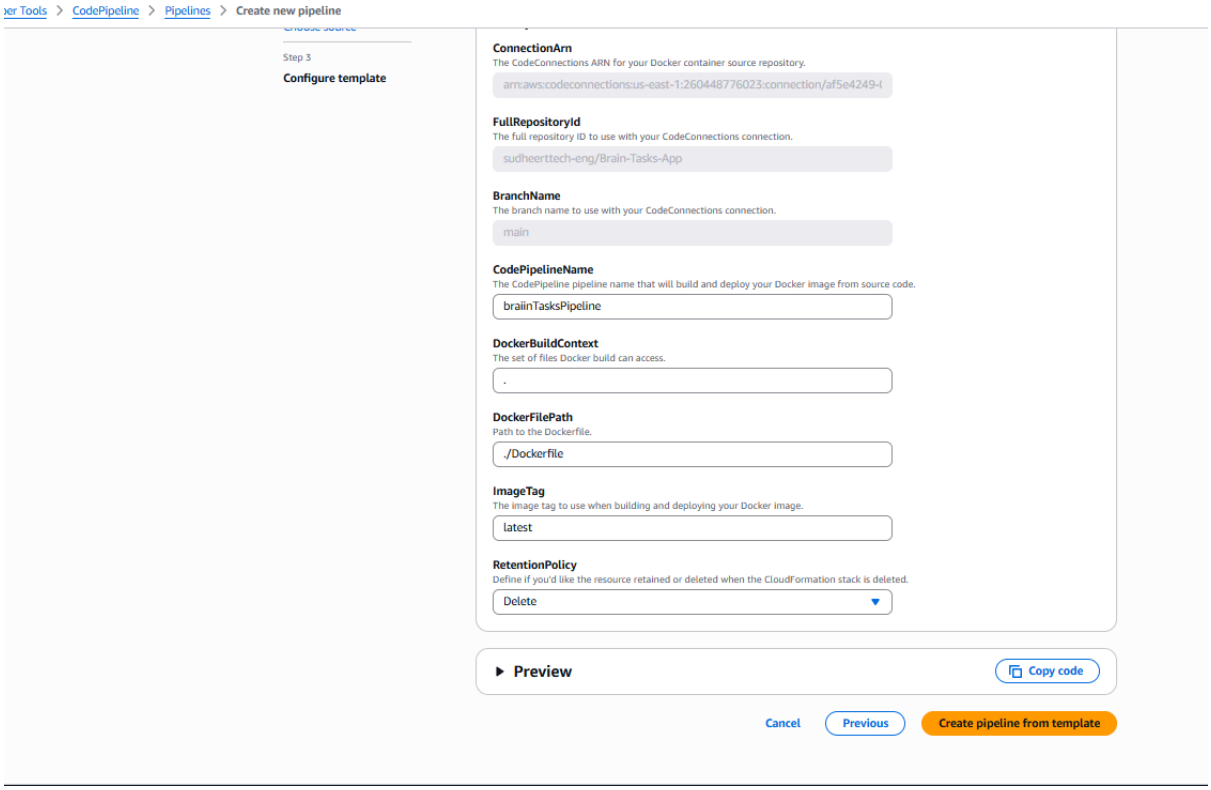
```
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ cd ..

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App (main)
$ git commit -m "moved files"
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        deployment.yaml
        service.yaml

nothing added to commit but untracked files present (use "git add" to track)

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App (main)
$ git add .

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   deployment.yaml
        new file:   service.yaml


hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App (main)
$ git commit -m "moved files"
[main f39fe13] moved files
 2 files changed, 31 insertions(+)
 create mode 100644 deployment.yaml
 create mode 100644 service.yaml

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App (main)
$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 696 bytes | 232.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sudheerttech-eng/Brain-Tasks-App.git
   3f2f909..f39fe13  main -> main
```

```
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App/dist (main)
$ cd ..

hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App (main)
$ ls
Dockerfile  README.md  buildspec.yml  deployment.yaml  dist/  service.yaml
```

CodePipeline:

Creating CodePipeline:





Create new pipeline

Step 2
Choose pipeline settings

Step 3
Add source stage

Step 4 - *optional*
Add build stage

Step 5 - *optional*
Add test stage

Step 6
Add deploy stage

Step 7
Review

Source

Source provider
This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

GitHub (via GitHub App)

Connection
Choose an existing connection that you have already configured, or create a new one and then return to this task.

arn:aws:codeconnections:us-

or Connect to GitHub

Repository name
Choose a repository in your GitHub account.

sudheerttech-eng/Brain-Tasks-App

You can type or paste the group path to any project that the provided credentials can access. Use the format 'group/subgroup/project'.

Default branch
Default branch will be used only when pipeline execution starts from a different source or manually started.

main

Output artifact format
Choose the output artifact format.

- CodePipeline default
  AWS CodePipeline uses the default zip format for artifacts in the pipeline. Does not include Git metadata about the repository.
- Full clone
  AWS CodePipeline passes metadata about the repository that allows subsequent actions to do a full Git clone. Only supported for AWS CodeBuild actions. Learn more

☑ Enable automatic retry on stage failure

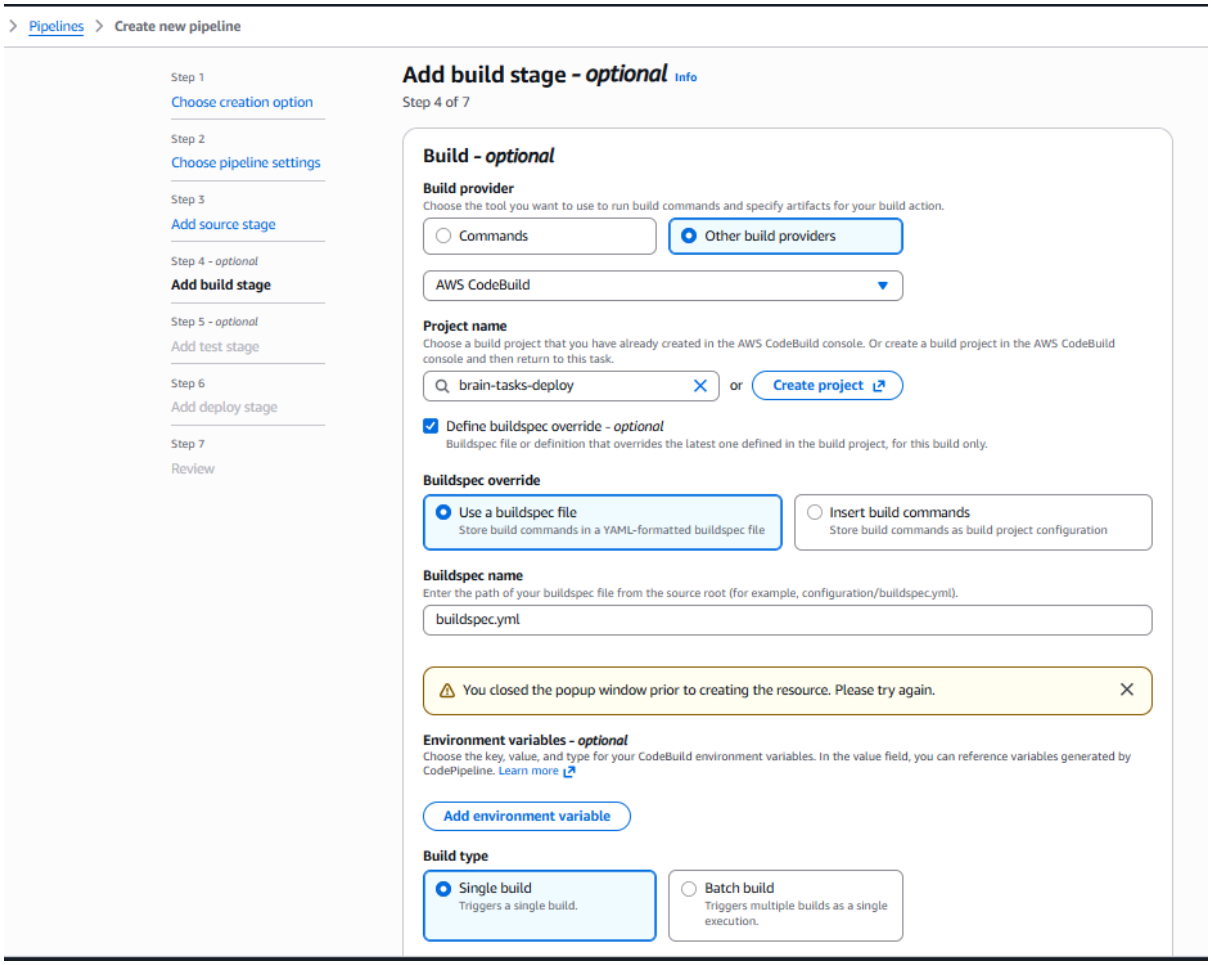
Webhook events

Webhook - *optional*

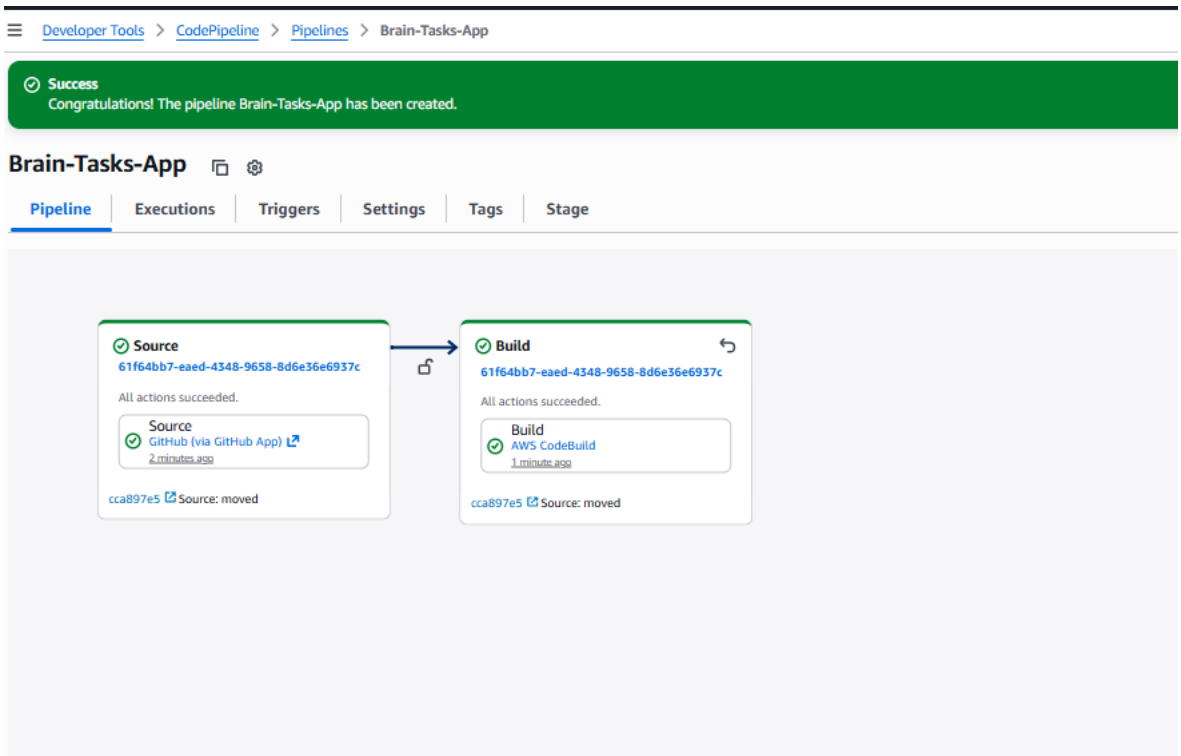
☑ Start your pipeline on push and pull request events.

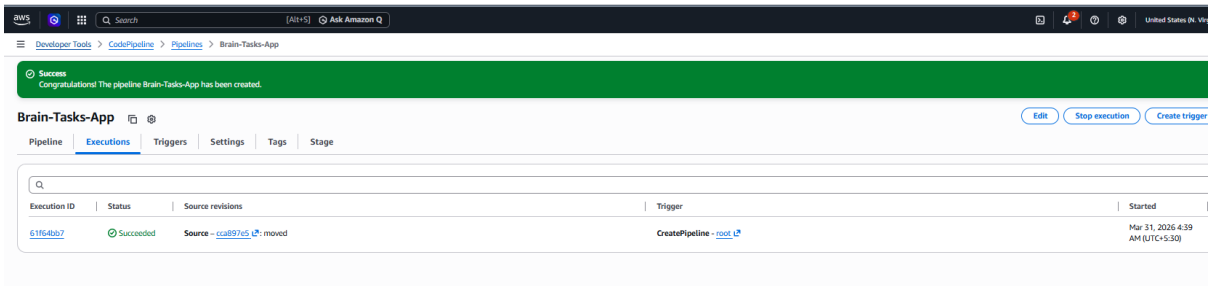
▸ **Webhook event filters - *optional***
Starts your pipeline on a specific event.

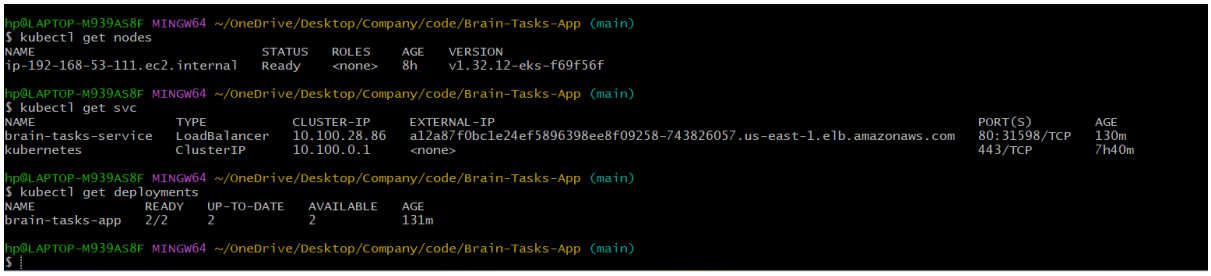
Remove filters

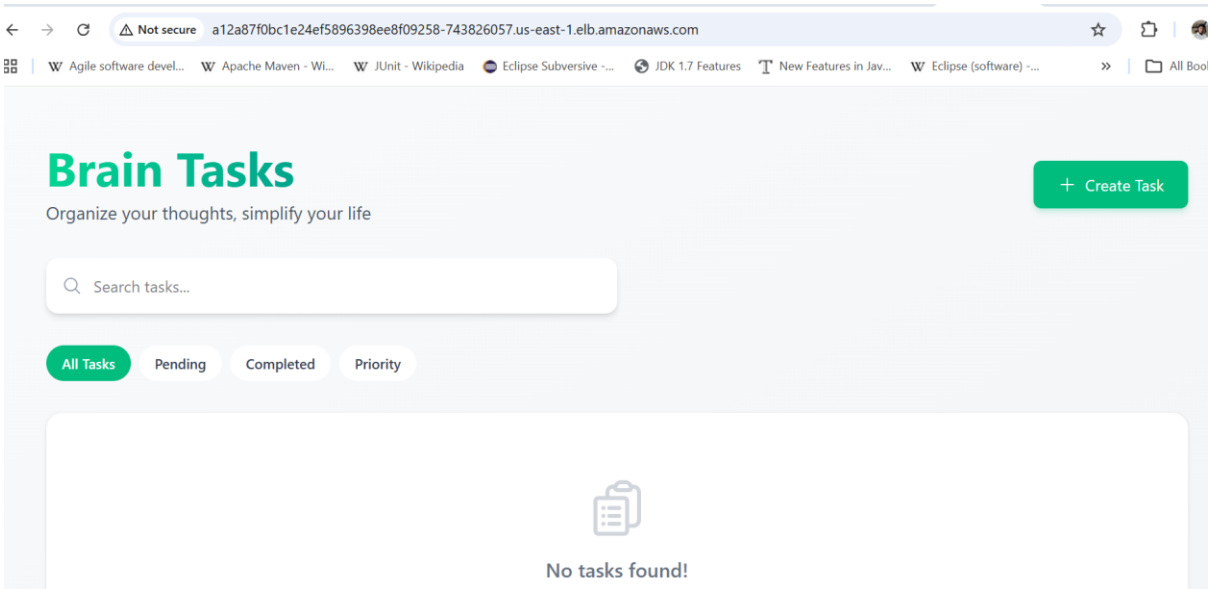


Deployed to EKS via Codepipeline EKS deploy stage..




Developer Tools > CodePipeline > Pipelines > Brain-Tasks-App
Success
Congratulations! The pipeline Brain-Tasks-App has been created.
Brain-Tasks-App
Edit
Stop execution
Create trigger
Pipeline
Executions
Triggers
Settings
Tags
Stage
Execution ID
Status
Source revisions
Trigger
Started
61f64bb7
Succeeded
Source – cca897e5: moved
CreatePipeline - root
Mar 31, 2026 4:39 AM (UTC+5:30)


hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App (main)
$ kubectl get nodes
NAME STATUS ROLES AGE VERSION
ip-192-168-53-111.ec2.internal Ready <none> 8h v1.32.12-eks-f69f56f
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App (main)
$ kubectl get svc
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
brain-tasks-service LoadBalancer 10.100.28.86 a12a87f0bc1e24ef5896398ee8f09258-743826057.us-east-1.elb.amazonaws.com 80:31598/TCP 130m
kubernetes ClusterIP 10.100.0.1 <none> 443/TCP 7h40m
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App (main)
$ kubectl get deployments
NAME READY UP-TO-DATE AVAILABLE AGE
brain-tasks-app 2/2 2 2 131m
hp@LAPTOP-M939AS8F MINGW64 ~/OneDrive/Desktop/Company/code/Brain-Tasks-App (main)
$


Not secure
a12a87f0bc1e24ef5896398ee8f09258-743826057.us-east-1.elb.amazonaws.com
Agile software devel...
Apache Maven - Wi...
JUnit - Wikipedia
Eclipse Subversive -...
JDK 1.7 Features
New Features in Jav...
Eclipse (software) -...
Brain Tasks
Organize your thoughts, simplify your life
+ Create Task
Search tasks...
All Tasks
Pending
Completed
Priority
No tasks found!

Monitoring:

Build history | Batch history

Build history

Stop build | View artifacts | View logs | Delete builds | Retry build | Del

Build run	Status	Project	Build number	Source version	Submitter	Duration	Completed
brain-tasks-deploy:7ecc3e93-e2e6-4bab-98c9-495493f1e461	Succeeded	brain-tasks-deploy	25	arn:aws:s3:::codepipeline-us-east-1-79d6bcb739f9-4907-9bd5-f91d1e80902d/Brain-Tasks-App/SourceArti/0f2OJgt	codepipeline/Brain-Tasks-App	44 seconds	9 minutes ago
SimpleDockerProject-0e349099ae8d:c4aa44ac-2f57-4de0-9375-a08d0b098120	Succeeded	SimpleDockerProject-0e349099ae8d	1	arn:aws:s3:::codepipelinestartertempla-codepipelineartifactsbuc-qckyOy5e8xby/braiinTasksPipeline/SourceOutp/UV8nrGh	codepipeline/braiinTasksPipeline	30 seconds	1 hour ago
brain-tasks-deploy:67f53e4b-bc9e-4fe1-8de7-af56052be73a	Succeeded	brain-tasks-deploy	24	-	root	50 seconds	2 hours ago
brain-tasks-deploy:319a5806-d80c-4bec-ae31-a61dcfa2b78a	Failed	brain-tasks-deploy	23	-	root	40 seconds	2 hours ago
brain-tasks-deploy:55fa4820-d6a3-4347-bf32-d18f6ccd515e	Failed	brain-tasks-deploy	22	-	root	41 seconds	2 hours ago
brain-tasks-deploy:7a65f653-ca03-4ccc-999c-3ff116c21d21e	Failed	brain-tasks-deploy	21	-	root	44 seconds	2 hours ago
brain-tasks-deploy:fa2617fc-039f-40fc-9a4d-7bac6464c7eb	Failed	brain-tasks-deploy	20	-	root	46 seconds	2 hours ago
brain-tasks-deploy:84fcf312-f88e-457a-bb24-ea1278b5a94d	Failed	brain-tasks-deploy	19	-	root	51 seconds	3 hours ago
brain-tasks-deploy:cedf1842-d2ee-4577-940d-3f306e9c22e6	Failed	brain-tasks-deploy	18	-	root	33 seconds	3 hours ago
brain-tasks-deploy:659d3681-2ec3-43f6-8680-8ac708077c00	Failed	brain-tasks-deploy	17	-	root	32 seconds	4 hours ago

Cloud watch logs:

CloudWatch > Log management

loudWatch

Log groups | Data sources - *new* | Summary

▸ Get-set-go with unified data store

Log groups (4)

By default, we only load up to 10,000 log groups.

Filter log groups or try pattern search | Exact match

Log group	Log class	Anomaly d...	Deletion protection	Bearer token authentic...	Data protection	Sensitive data count	Retention
/aws/cloudtrail/s3-upload-logs	Standard	Configure	Off	Off	-	-	Never expire
/aws/codebuild/SimpleDockerProject-0e349099ae8d	Standard	Configure	Off	Off	-	-	Never expire
/aws/codebuild/brain-tasks-deploy	Standard	Configure	Off	Off	-	-	Never expire
aws-cloudtrail-logs-260448776023-7a08e472	Standard	Configure	Off	Off	-	-	Never expire

CloudWatch > Log management > /aws/codebuild/brain-tasks-deploy

Log streams | Tags | Data protection | Anomaly detection | Metric filters | Subscription filters | Contributor Insights | Field indexes | Transformer

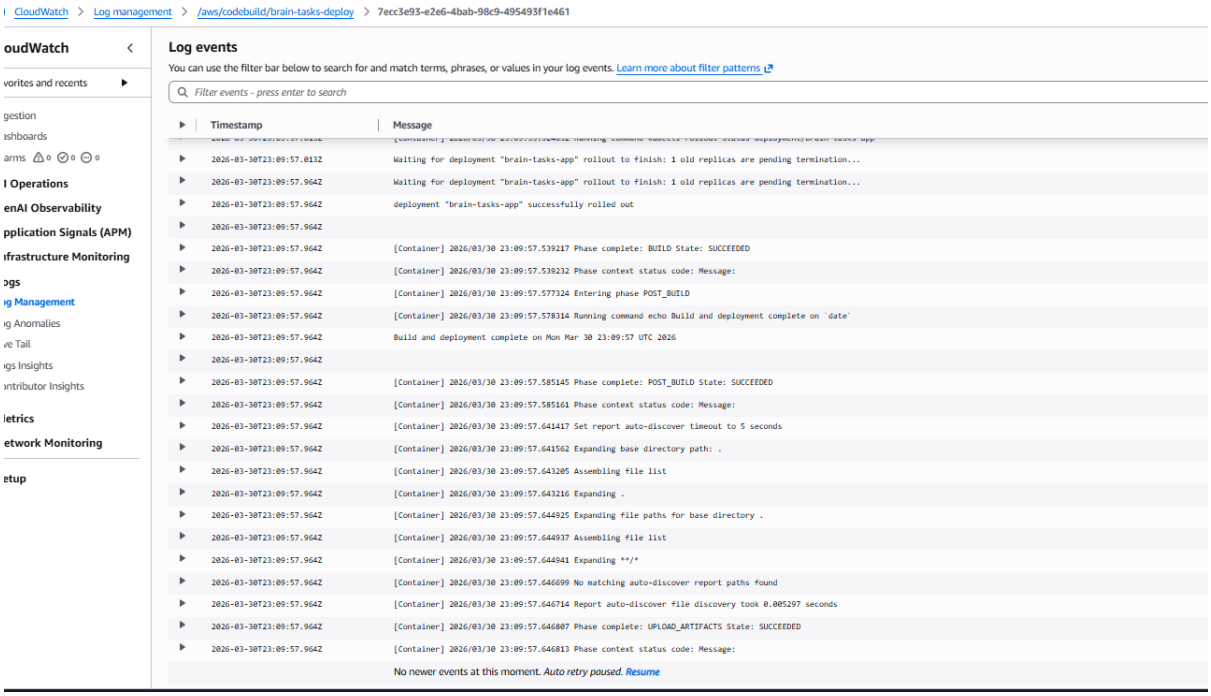
Log streams (25)

By default, we only load the most recent log streams.

Filter log streams or try prefix search | Exact match | Show expired | Info

Delete | Create log stream

Log stream	Last event time
7ecc3e93-e2e6-4bab-98c9-495493f1e461	2026-03-30 23:09:28 (UTC)
67f53e4b-bc9e-4fe1-8de7-af56052be73a	2026-03-30 21:04:57 (UTC)
319a5806-d80c-4bec-ae31-a61dcfa2b78a	2026-03-30 20:56:07 (UTC)
55fa4820-d6a3-4347-bf32-d18f6ccd515e	2026-03-30 20:49:59 (UTC)
7a65f653-ca03-4ccc-999c-3ff116c21d21e	2026-03-30 20:42:18 (UTC)
fa2617fc-039f-40fc-9a4d-7bac6464c7eb	2026-03-30 20:29:08 (UTC)
84fcf312-f88e-457a-bb24-ea1278b5a94d	2026-03-30 19:34:09 (UTC)
cedf1842-d2ee-4577-940d-3f306e9c22e6	2026-03-30 19:24:26 (UTC)
659d3681-2ec3-43f6-8680-8ac708077c00	2026-03-30 19:19:06 (UTC)
77835ec5-a5da-475e-8c04-dff1cdc96010	2026-03-30 19:11:01 (UTC)
7d1ca621-a311-4af6-8f50-7ee7dfdf64a1	2026-03-30 18:51:40 (UTC)
7bbd0971-3dd7-4c56-8641-1c3e865f8960	2026-03-30 18:44:55 (UTC)
9b04251d-20b2-4b4f-a290-a165782bc6e3	2026-03-30 18:38:20 (UTC)
4a34c99f-6b97-472f-ba28-7364babb377b	2026-03-30 17:44:25 (UTC)
7f116c12-a698-43ee-90d2-8a89af44ef7b	2026-03-30 17:39:15 (UTC)



Load Balancer ARN: http://a12a87f0bc1e24ef5896398ee8f09258-743826057.us-east-1.elb.amazonaws.com/